Bar chain models, E_1-chiral quantum groups, and the missing operad

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Abstract

We study the bar complex of a chiral algebra as a factorisation coalgebra and examine the chain-level structures it carries beyond its cohomology. Several imprecisions in the standard informal treatment are identified and corrected: the bar complex is not factorisation homology of the line but of the interval with boundary conditions; the E_2 structure relevant to 3d holomorphic-topological field theory lives on the module category, not on the algebra; and the r-matrix visible in the bar complex is the classical shadow of a quantum group structure that requires a chiral coproduct as independent input. We formulate the definition of an E_1-chiral quantum group, identify the modular operad governing A_∞-algebras in E_1-chiral algebras as the missing structural ingredient connecting the bar complex programme to 3d quantum gravity, and pose the open questions whose resolution would complete the connection.

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1 Introduction

The bar complex $B(\mathcal{A})$ of a chiral algebra $\mathcal{A}$ on a smooth curve X is a factorisation coalgebra on the Ran space $\mathrm{Ran}(X)$. Its derived global sections compute chiral homology; its bar filtration produces a spectral sequence whose degeneration detects Koszul duality; and its genus-g curvature $d_B^2 = \kappa_{\mathrm{ch}}\,\omega_g$ extracts a tower of numerical invariants (the shadow obstruction tower) that govern the modular properties of the chiral algebra.

These constructions, developed across two volumes [1, 2], provide compelling evidence for a conjecture: that the bar complex of a boundary chiral algebra encodes a 3-dimensional quantum gravity theory on $X \times \mathbb{R}_{\geq 0}$. The present note examines this conjecture critically, identifies several imprecisions in its informal formulation, and isolates the structural gap that separates the existing results from a proof.

The gap is the absence of a modular operad governing A_∞-algebras in E_1-chiral algebras. The bar complex provides the invariants; this operad would provide the objects producing them.

Conventions. We work over $\mathbb{C}$. "Chiral algebra" means factorisation algebra on a smooth curve in the sense of Beilinson–Drinfeld. "E_1-chiral algebra" means an E_1-algebra object in chiral algebras, as defined in [2]. All operadic structures are in the ∞-categorical sense unless stated otherwise.

2 The bar complex as factorisation coalgebra

2.1 What the bar complex is

Let $\mathcal{A}$ be a chiral algebra on a smooth curve X. The bar construction

$$B(\mathcal{A}) = \bigoplus_{n\geq 0} (s\mathcal{A})^{\otimes n}, \qquad d_B = \sum_{k\geq 2} \sum_i m_k^{(i)},$$

equipped with the deconcatenation coproduct

$$\Delta(a_1|\cdots|a_n) = \sum_{i=0}^{n} (a_1|\cdots|a_i) \otimes (a_{i+1}|\cdots|a_n),$$

is a dg coalgebra. Here $m_k^{(i)}$ applies the A_∞-operation m_k to a consecutive block of k tensor factors starting at position i, and s denotes the bar suspension.

Theorem 2.1 (Vol. I, Theorem A)**.** *$B(\mathcal{A})$ carries the structure of a factorisation coalgebra on* $\mathrm{Ran}(X)$*. The cobar construction Ω provides a left adjoint, and $\Omega B(\mathcal{A}) \simeq \mathcal{A}$ for connected $\mathcal{A}$.*

The bar complex is a *local* object on $\mathrm{Ran}(X)$. Chiral homology $H_*^{\mathrm{ch}}(X, \mathcal{A})$ is the *global* invariant obtained by taking derived global sections of this factorisation coalgebra. The bar complex is to chiral homology as a sheaf is to its cohomology.

2.2 What the bar complex is not

Warning 2.2. The bar complex is not factorisation homology of the line. For an E_1-algebra A, factorisation homology of $\mathbb{R}$ is A itself: $\int_{\mathbb{R}} A \simeq A$. The bar construction computes a different invariant:

$$B(A) \;\simeq\; k \otimes_A^{\mathbf{L}} k \;\simeq\; \int_{[0,1]}^{k,\,k} A,$$

the factorisation homology of the **compact interval** with augmentation boundary conditions at both endpoints. The boundary conditions are essential; without them, the answer is trivially A.

2.3 The three bar complexes

For a vertex algebra V (an E_∞-chiral algebra), three bar-type constructions are available. They are three different functors applied to three different algebraic structures carried by V.

(i) $B^{FG}(V)$: the Francis–Gaitsgory bar complex, using only the commutative product (the normally-ordered product, ignoring OPE poles). This is the factorisation homology of the underlying commutative algebra.

(ii) $B^{\mathrm{ord}}(V)$: the ordered bar complex, using the full OPE with tensor factors in a fixed sequence. The A_∞-operations m_k encode the OPE pole data after $d\log$-extraction: an OPE pole of order p contributes to m_p.

(iii) $B^{\Sigma}(V)$: the symmetric bar complex, using the full OPE with Σ_n-coinvariants. This is the chiral bar complex whose derived global sections compute chiral homology.

The descent from ordered to symmetric,

$$B_n^{\mathrm{ord}} \twoheadrightarrow B_n^{\Sigma} = \left(B_n^{\mathrm{ord}}\right)^{R\text{-}\Sigma_n},$$

is R-matrix twisted: the transposition $\sigma_{i,i+1}$ acts via the braiding $R(z_i - z_{i+1})$ derived from analytic continuation of the OPE. The Yang–Baxter equation ensures coherence. For E_∞-chiral algebras, this R-matrix is canonically determined by the OPE.

Remark 2.3. The R-twisted coinvariants formula is schematic. The R-matrix $R(z)$ depends on a spectral parameter $z = z_i - z_{i+1}$, which parametrises positions on the configuration space. The symmetrisation acts on the integrand (before integration over configuration space), not on the bar complex (after integration). The formula $B_n^{\Sigma} = (B_n^{\mathrm{ord}})^{R\text{-}\Sigma_n}$ is shorthand for: twist the configuration-space integrand by the R-matrix action, then integrate.

3 Bar chain models for stratified 1-configurations

Bar chain models are indexed not by manifolds but by *stratified configurations*: topologico-combinatoric stratifications of 1-dimensional spaces equipped with bulk-to-boundary structure. The bar complex $B(\mathcal{A}) = k \otimes^{\mathbf{L}}_{\mathcal{A}} k$ is the bar chain model for the stratified configuration of the compact interval with two boundary strata and augmentation data at each. Other stratified configurations—varying the combinatorics of bulk and boundary strata—give other bar chain models, each computing a different invariant.

3.1 $[0,1]$ with two boundary points, augmentation

For an E_1-algebra A with left module M and right module N:

$$\int_{[0,1]}^{M,N} A = M \otimes^{\mathbf{L}}_{A} N.$$

With $M = N = k$ (augmentation): this is $B(A)$, the standard bar construction. The chain model uses ordered configurations on $[0,1]$ compactified by the associahedra K_n. The deconcatenation coproduct reflects the ability to cut the interval at any interior point. This is the Koszul dual; it encodes the representation theory of A.

3.2 S^1 with no boundary, cyclic identification

$$\int_{S^1} A = \mathrm{HH}_*(A, A),$$

the Hochschild homology of A with coefficients in itself. The chain model is the cyclic bar complex: configurations of n ordered points on S^1, with wraparound—the last point can collide with the first. For an E_1-chiral algebra, the wraparound involves the full R-matrix monodromy around the circle.

This bar chain model is where the derived centre lives. The derived centre $Z(\mathcal{A}) = \mathrm{HH}^*(\mathcal{A}, \mathcal{A})$, computed by the Hochschild cochain complex, is the endomorphism algebra

of the identity functor on $\mathrm{Mod}_{\mathcal{A}}$. Its E_2 structure (by the Deligne conjecture) governs the holomorphic-topological bulk. The genus-1 curvature $d_B^2 = \kappa_{\mathrm{ch}}\,\omega_1$ is a chain-level obstruction visible in this bar chain model.

3.3 $[0,\infty)$ with one boundary point, module datum

For a right A-module M:

$$\int_{[0,\infty)}^{M} A \;\simeq\; M.$$

The half-line is contractible, so factorisation homology collapses to the boundary datum. The chain model is the one-sided bar resolution $\cdots \to A^{\otimes 2} \otimes M \to A \otimes M \to M$, which resolves M as an A-module. The chain-level data detects the A_∞-module structure and provides the bar filtration on modules.

3.4 $\mathbb{H}$ with boundary $\mathbb{R}$, bulk-to-boundary Swiss-cheese stratification

The upper half-plane $\mathbb{H}$ with boundary $\mathbb{R} = \partial\mathbb{H}$ is the geometry of the Swiss-cheese operad. Its bar chain model involves two-dimensional configurations: m bulk points in $\mathbb{H}$ and n boundary points on $\mathbb{R}$, with three types of collision (bulk–bulk, boundary–boundary, bulk–boundary). This model encodes the $\mathrm{SC}^{\mathrm{ch,top}}$ structure at the chain level: how chiral data on $\mathbb{C}$ interacts with ordered data on the boundary.

4 Two A_∞ structures and Koszul duality

The connection between the bar complex of a chiral algebra and 3-dimensional holomorphic-topological (HT) field theory passes through two A_∞ structures that are Koszul dual to each other.

4.1 The boundary A_∞-coalgebra

The bar complex $B(\mathcal{A})$ of the boundary chiral algebra $\mathcal{A}$ is an A_∞-coalgebra. The A_∞-coalgebra structure consists of:

- the deconcatenation coproduct Δ (structural, exists for any bar complex),
- the bar differential $d_B = \sum m_k$ (encoding the OPE poles of $\mathcal{A}$).

This A_∞-coalgebra lives on $\mathrm{Ran}(X)$ and is holomorphic in the X-direction.

4.2 The bulk A_∞-algebra

The Hochschild cochain complex

$$C^*(\mathcal{A},\mathcal{A}) = \mathrm{RHom}_{\mathcal{A}\text{-}\mathcal{A}}(\mathcal{A},\mathcal{A}) \simeq \mathrm{Hom}(B(\mathcal{A}),\mathcal{A})$$

carries a natural A_∞-algebra structure, dual to the A_∞-coalgebra structure on $B(\mathcal{A})$ through the Hom functor. By the Deligne conjecture (proved by Kontsevich, Tamarkin, and others), this extends to a full E_2 structure.

This E_2 structure decomposes, by Dunn additivity, as $E_1^{\mathrm{hol}} \otimes E_1^{\mathrm{top}}$:

- E_1^{hol}: the chiral algebra structure of $Z(\mathcal{A}) = \mathrm{HH}^*(\mathcal{A})$ on X,
- E_1^{top}: the A_∞-algebra structure in the transverse direction.

This is the A_∞-chiral algebra of the HT physicists: a chiral algebra on X with A_∞ structure in a transverse topological direction. It is the factorisation algebra of the 3d HT bulk on $X \times \mathbb{R}$.

4.3 Koszul duality bridges boundary and bulk

The two A_∞ structures—coalgebra on $B(\mathcal{A})$ and algebra on $C^*(\mathcal{A}, \mathcal{A})$—are Koszul dual through the Hom functor:

$$\underbrace{\mathcal{A}}_{\substack{\text{boundary}\\(E_1)}} \xrightarrow{B} \underbrace{B(\mathcal{A})}_{\substack{A_\infty\text{-coalgebra}\\\text{(from OPE poles)}}} \xrightarrow{\mathrm{Hom}(-,\mathcal{A})} \underbrace{C^*(\mathcal{A},\mathcal{A})}_{\substack{A_\infty\text{-algebra}\\\text{(transverse)}}} \xleftarrow{\sim} \underbrace{Z(\mathcal{A})}_{\substack{\text{bulk}\\(E_2)}}$$

The boundary chiral algebra is E_1. The bulk algebra (the derived centre) is E_2. The bar complex is the bridge: its A_∞-coalgebra structure dualises to the A_∞-algebra structure of the bulk. The Dunn decomposition $E_2 = E_1 \otimes E_1$ applies to $Z(\mathcal{A})$ (the bulk), not to $\mathcal{A}$ (the boundary).

Warning 4.1. The Dunn decomposition does *not* apply at the algebra level. The boundary algebra $\mathcal{A}$ is E_1 (associative, non-commutative). The E_2 structure lives on $Z(\mathcal{A})$ or equivalently on $\mathrm{Mod}_{\mathcal{A}}$. Applying Dunn to $\mathcal{A}$ itself is a categorical level error.

5 The categorical level shift

5.1 Quantum groups are E_1, not E_2

A quantum group $U_q(\mathfrak{g})$ is an E_1-algebra: its multiplication is associative and non-commutative. It is not E_2; it does not have homotopy-commutative multiplication.

What is E_2 is the *category of representations* $\mathrm{Rep}(U_q(\mathfrak{g}))$, which is a braided monoidal category—an E_2-algebra object in Cat. The R-matrix $R \in U_q(\mathfrak{g})^{\otimes 2}$ provides the braiding $V \otimes W \xrightarrow{\sim} W \otimes V$ on modules. It acts on modules, not on the algebra.

Level	**Object**	**Structure**
Algebra	$U_q(\mathfrak{g})$	E_1
Module category	$\mathrm{Rep}(U_q(\mathfrak{g}))$	E_2 in Cat
Centre	$Z(U_q(\mathfrak{g}))$	E_2

5.2 The level shift for chiral algebras

The same level shift governs the chiral setting:

Level	**Object**	**Structure**
Boundary algebra	$\mathcal{A}$	E_1-chiral
Module category	$\mathrm{Mod}_{\mathcal{A}}$	E_2 in Cat (if braided)
Derived centre	$Z(\mathcal{A}) = \mathrm{HH}^*(\mathcal{A})$	E_2-chiral (the HT bulk)

The E_2 structure relevant to 3d holomorphic-topological field theory lives at the level of the derived centre $Z(\mathcal{A})$, not at the level of $\mathcal{A}$ itself. The Dunn decomposition $E_2 = E_1^{\mathrm{hol}} \otimes E_1^{\mathrm{top}}$ applies to $Z(\mathcal{A})$ and gives the holomorphic + topological structure of the 3d bulk.

5.3 The chiral/topological distinction

At each E_N level, one must distinguish *chiral* from *topological* structure. The distinction is whether the algebraic data depends on the complex structure of the surface X.

Definition 5.1. *An E_N-**chiral** structure on $X \times \mathbb{R}^{N-1}$ is an E_N-algebra in factorisation algebras that depends on the complex structure of X. An E_N-**topological** structure is one that is independent of the complex structure: it depends only on the smooth (or topological) type of the underlying manifold.*

The E_2 structure on the derived centre $Z(\mathcal{A}) = \mathrm{HH}^*(\mathcal{A})$ discussed above is E_2-*chiral*: the Dunn decomposition $E_2 = E_1^{\mathrm{hol}} \otimes E_1^{\mathrm{top}}$ has one holomorphic factor that depends on the complex structure of X. This is the HT (holomorphic $\times$ topological) structure of Costello and collaborators.

When the boundary algebra $\mathcal{A}$ carries a **conformal vector** (Sugawara construction), the holomorphic direction can be *topologised*: the stress tensor $T(z)$ generates conformal transformations that trivialise the dependence on complex structure. The E_2-chiral structure on $Z(\mathcal{A})$ then enhances to E_2-topological: a full 2d TQFT structure independent of the complex structure of X.

More generally, the climax of Volume II concerns the E_3 level. The derived centre $Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$, as a 3d bulk theory on $X \times \mathbb{R}_{\geq 0}$, is naturally E_3-chiral (holomorphic on X, topological on $\mathbb{R}^2$). When $\mathcal{A}$ has conformal vector at non-critical level, topologisation promotes this to E_3-topological: a full 3d TQFT, i.e. Chern–Simons theory.

Warning 5.2. Topologisation fails at the **critical level** $k = k_{\mathrm{crit}}$. At critical level, the Sugawara construction degenerates: the conformal vector ceases to exist. The E_3-chiral structure persists but cannot be promoted to E_3-topological. This is the algebraic shadow of the fact that Chern–Simons theory at critical level (equivalently, level 0 after the standard shift) is degenerate.

The following table summarises the distinction across categorical levels:

Level	**Object**	**Structure**	**Type**
Boundary algebra	$\mathcal{A}$	E_1-chiral	chiral
Module category	$\mathrm{Mod}_{\mathcal{A}}$	E_2 in Cat	chiral
Derived centre	$Z(\mathcal{A}) = \mathrm{HH}^*(\mathcal{A})$	E_2-chiral	chiral (HT)
+ conformal vector	$Z(\mathcal{A})$	E_2-topological	topological (TQFT)
3d bulk	$Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$	E_3-chiral	chiral (HT)
+ conformal vector	$Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$	E_3-topological	topological (CS)

The passage from the penultimate row to the last—from E_3-chiral to E_3-topological—is the climax of Volume II. It is the conformal vector that provides the bridge, and the critical level that marks its boundary.

6 E_1-chiral quantum groups

6.1 Definition

Definition 6.1. *An E_1-**chiral quantum group** on a curve X is an E_1-chiral algebra $\mathcal{A}$ on X equipped with:*

*(a) a **chiral coproduct** $\Delta : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$ (a coassociative coalgebra structure on $\mathcal{A}$ in the appropriate tensor product of chiral algebras),*

*(b) an R-**matrix** $R(z) \in \mathcal{A}^{\hat{\otimes} 2}$,*

*(c) **quasi-triangularity**: $R\,\Delta(a) = \Delta^{\mathrm{op}}(a)\,R$ for all a,*

*(d) an **antipode** $S : \mathcal{A} \to \mathcal{A}$ satisfying the Hopf compatibility,*

such that the category of chiral $\mathcal{A}$-modules $\mathrm{Mod}_{\mathcal{A}}$ is a braided monoidal category (an E_2-algebra in Cat).

Remark 6.2. The chiral coproduct (a) is structure on $\mathcal{A}$ itself. It is *not* the deconcatenation coproduct on $B(\mathcal{A})$. The deconcatenation coproduct is a structural feature of the bar construction, present for any algebra. The chiral coproduct is a Hopf-type structure that enables tensor products of $\mathcal{A}$-modules. Without it, $\mathrm{Mod}_{\mathcal{A}}$ is a category but has no monoidal structure, and there is nothing to braid.

6.2 What the bar complex provides, and what it does not

The bar complex $B(\mathcal{A})$ of an E_1-chiral algebra provides:

- the A_∞-coalgebra structure (bar differential + deconcatenation),
- the classical r-matrix $r(z) = k\,\Omega/z$ (from collision residues, after $d$log-extraction),
- the curvature $d_B^2 = \kappa_{\mathrm{ch}}\,\omega_g$ (the genus tower),
- the Koszul dual algebra $\mathcal{A}^!$ (from bar cohomology at genus 0).

The bar complex does *not* provide:

- the chiral coproduct $\Delta : \mathcal{A} \to \mathcal{A} \otimes \mathcal{A}$,
- the full quantum R-matrix $R(z)$ (only its classical limit $r(z)$),
- the quasi-triangularity condition,
- the antipode.

The r-matrix is necessary but not sufficient for a chiral quantum group. It gives the braiding data semiclassically—what the $\mathrm{Mod}_{\mathcal{A}}$ braiding should be in the classical limit—but without the chiral coproduct on $\mathcal{A}$ there is no monoidal structure on $\mathrm{Mod}_{\mathcal{A}}$ and therefore nothing to braid.

6.3 The coproduct is not visible in the shadows

The shadow obstruction tower $\{F_g\}_{g\geq 1}$ extracts numerical invariants from the curvature of the bar complex. These invariants detect collision and fusion data: how operators come together. Fusion is the product.

The coproduct encodes the opposite operation: how a single operator splits into a pair. Product and coproduct are independent algebraic structures. The shadow tower, built from the bar differential (which encodes the product/OPE), does not see the coproduct.

6.4 Sources of the chiral coproduct

Two recent constructions provide chiral/vertex coproducts:

(i) **Gaiotto–Rapčák–Zhou** [3]: coproducts on the deformed double current algebra of $\mathfrak{gl}_K$ and the $\mathfrak{gl}_K$-extended $\mathcal{W}_\infty$, constructed from M2–M5 brane fusions in Costello's twisted M-theory.

(ii) **Jindal–Kaubrys–Latyntsev** [4]: a vertex coproduct on the Kontsevich–Soibelman cohomological Hall algebra (CoHA) of a quiver with potential, forming a vertex bialgebra. For ADE quivers, this recovers Drinfeld's deformed coproduct on the Yangian.

Question 6.3. *Do these coproducts lift to chiral coproducts on Kac–Moody vertex algebras $V_k(\mathfrak{g})$ for arbitrary simple $\mathfrak{g}$? The Gaiotto–Rapčák–Zhou construction is for type A ($\mathfrak{gl}_K$). The CoHA construction works for ADE quivers. Whether either extends to give a canonical chiral coproduct on $V_k(\mathfrak{g})$ itself (rather than on associated Yangians or CoHAs) is open.*

7 The missing operad

7.1 The structural gap

The existing programme constructs:

(1) E_1-chiral algebras on X (the boundary objects),

(2) their bar complexes $B(\mathcal{A})$ (factorisation coalgebras on $\mathrm{Ran}(X)$),

(3) the classical r-matrices (from collision residues),

(4) the curvature $d_B^2 = \kappa_{\mathrm{ch}}\,\omega_g$ (the shadow tower),

(5) the Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ (governing the within-surface holomorphic-topological interaction).

What the programme does *not* construct is the **modular operad governing A_∞-algebras in E_1-chiral algebras**. This is the operad whose algebras are the 3d HT bulk theories with E_1-chiral boundary.

7.2 What this operad must encode

An algebra over this operad should consist of:

(a) an E_1-chiral algebra structure on X (the boundary/holomorphic direction: ordered operations on the curve, with OPE poles),

(b) an A_∞-algebra structure in a transverse direction $\mathbb{R}$ (the bulk/topological direction),

(c) compatibility data: how the transverse A_∞-operations interact with the within-X E_1 ordering,

(d) at higher genus: a modular extension, with curvature $\kappa_{\mathrm{ch}}\,\omega_g$ as the obstruction to extending from genus 0.

$\mathrm{SC}^{\mathrm{ch,top}}$ encodes the within-surface structure (a). The full operad would combine (a) with the transverse A_∞ structure (b) and the compatibility data (c).

Remark 7.1 (On two E_1 structures). An algebra over this operad carries two E_1 structures:

- $E_1^{(1)}$: the E_1-chiral algebra structure on X (from (a)),
- $E_1^{(2)}$: the A_∞-algebra structure on $\mathbb{R}$ (from (b)).

These are *not* independent. They are Koszul dual through the bar construction (Section 4). The compatibility datum (c) encodes this Koszul duality at the operadic level.

For an E_∞-chiral algebra, $E_1^{(1)}$ is derived from the local OPE and does not add independent information. For a genuinely E_1-chiral algebra, $E_1^{(1)}$ is independent input—the ordered collision data is not determined by any local OPE—and the compatibility with $E_1^{(2)}$ is a non-trivial constraint.

Remark 7.2 (Relation to chiral quantum groups). The chiral coproduct of Definition 6.1 should arise as part of the structure of an algebra over this operad. The operad would provide the framework in which the coproduct, the R-matrix, and the transverse A_∞ structure cohere into a single algebraic object. The R-matrix would not be imposed but would emerge as the coherence datum between $E_1^{(1)}$ and $E_1^{(2)}$—visible at the level of the module category $\mathrm{Mod}_{\mathcal{A}}$ (which is E_2 in Cat), not at the level of $\mathcal{A}$ itself (which remains E_1).

7.3 Curved extensions

At genus $g \geq 1$, the bar complex has $d_B^2 = \kappa_{\mathrm{ch}}\,\omega_g \neq 0$. The A_∞ structure is *curved*: the Stasheff relations are modified by the curvature term. Standard Dunn additivity applies to E_n-algebras, not to curved algebras.

Conjecture 7.3 (Curved Dunn additivity)**.** *There exists a curved analogue of the Dunn decomposition: a curved E_2-algebra decomposes as a curved E_1-algebra in curved E_1-algebras, with curvature $\kappa_{\mathrm{ch}}\,\omega_g$ as the obstruction to the flat decomposition.*

A proof would extend the chiral quantum group framework to higher genus and connect the shadow obstruction tower to curved operadic algebra.

8 The 3d gravity conjecture: status

8.1 What is proved

(1) **Theorem A** (bar–cobar adjunction): $B(\mathcal{A})$ is a factorisation coalgebra, $\Omega B(\mathcal{A}) \simeq \mathcal{A}$.

(2) **Hochschild reconstruction**: the derived centre $Z(\mathcal{A}) = \mathrm{HH}^*(\mathcal{A})$ is a chiral algebra on X carrying an E_2 structure. This is proved (boundary-linear bulk reconstruction).

(3) **Perturbative CS match**: at genus 0, the bar complex of the Kac–Moody vertex algebra reproduces perturbative Chern–Simons theory (via the Costello–Francis–Gwilliam programme).

(4) **Curvature at genus 1**: $d_B^2 = \kappa_{\mathrm{ch}}\,\omega_1$ matches the one-loop gravitational anomaly.

(5) **Shadow tower**: the genus-g invariants F_g match Faber–Pandharipande intersection numbers on $\overline{\mathcal{M}}_g$.

8.2 What is conjectured

Conjecture 8.1 (3d gravity = derived centre)**.** *The derived centre $Z(\mathcal{A})$, with its E_2 structure,* is *the factorisation algebra of observables of 3d quantum gravity on $X \times \mathbb{R}_{\geq 0}$ with boundary theory $\mathcal{A}$.*

This requires:

- identifying the abstract Hochschild E_2 structure with the BV-quantised HT theory (beyond the perturbative match),
- extending the identification to all genera (the full global triangle boundary $\to$ bulk $\to$ boundary, not just the boundary-linear level),
- showing the bar complex satisfies surgery/gluing axioms in the transverse direction (giving invariants of 3-manifolds, not just of $X \times \mathbb{R}_{\geq 0}$).

8.3 What is missing

The **modular operad governing A_∞-algebras in E_1-chiral algebras** (Section 7). The bar complex provides the invariants. This operad would provide the objects producing them. Without it, the 3d gravity conjecture is a collection of matching invariants, not a structural identification.

9 Open questions

Q1. Construct the modular operad. Define the modular operad whose algebras are A_∞-algebras in E_1-chiral algebras on X. Its genus-0 part should recover the existing bar complex / SC structure. Its higher-genus part should produce the curvature $\kappa_{\mathrm{ch}}\,\omega_g$ as the operadic obstruction.

Q2. Construct the chiral coproduct for Kac–Moody. Determine whether the Gaiotto–Rapčák–Zhou coproducts (type A, from M2–M5 fusions) or the Jindal–Kaubrys–Latyntsev vertex coproducts (ADE quivers, from CoHA) lift to chiral coproducts on $V_k(\mathfrak{g})$ for arbitrary simple $\mathfrak{g}$.

Q3. Prove curved Dunn additivity. Establish Conjecture 7.3: the curved Dunn decomposition for curved E_2-algebras, with curvature $\kappa_{\mathrm{ch}}\,\omega_g$.

Q4. Construct genuinely E_1-chiral quantum groups. Find an E_1-chiral algebra $\mathcal{A}$ (not E_∞, not derived from any vertex algebra) equipped with a chiral coproduct Δ and R-matrix $R(z)$ making $\mathrm{Mod}_{\mathcal{A}}$ braided monoidal.

Q5. Identify the correct monoidal structure. The construction "A_∞-algebras in E_1-chiral algebras" requires a symmetric monoidal structure on the category of E_1-chiral algebras. Determine whether the $*$-tensor product, the !-tensor product, or another structure is the correct choice.

Q6. Relate the spectral $R(z)$ to the categorical braiding. The E_2 braiding from Dunn additivity (on $Z(\mathcal{A})$) is a categorical braiding with no spectral parameter. The chiral R-matrix $R(z)$ depends on z. Show that the categorical braiding, applied in the $\mathcal{D}$-module enriched setting, naturally acquires position-dependence and recovers $R(z)$.

Q7. Prove the global triangle. Extend the boundary-linear bulk reconstruction ($\mathcal{B}_{\mathrm{bulk}} \to Z(\mathcal{A})$) to the full global triangle: boundary $\to$ bulk $\to$ boundary. This is conjectural beyond the Koszul locus.

Q8. Construct E_3-topological structure from conformal vector. Let $\mathcal{A}$ be an E_1-chiral algebra with conformal vector at non-critical level $k \neq k_{\mathrm{crit}}$. The derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ carries E_3-chiral structure (Section 5.3). Show that the conformal vector defines a topologisation functor

$$E_3\text{-chiral} \longrightarrow E_3\text{-topological}$$

and that the resulting E_3-topological theory on $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ recovers Chern–Simons theory (for $\mathcal{A} = V_k(\mathfrak{g})$, at the appropriate level). Determine the precise obstruction at critical level: is the failure of topologisation controlled by a class in the deformation complex of E_3-algebras?

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